

BAAZ & UDAY JKBOSE

2026 CLASS 11TH

CHEMICAL COORDINATION & INTEGRATION

LECTURE-05

ZOOLOGY

DAVOOD SIR (CEO of NG)

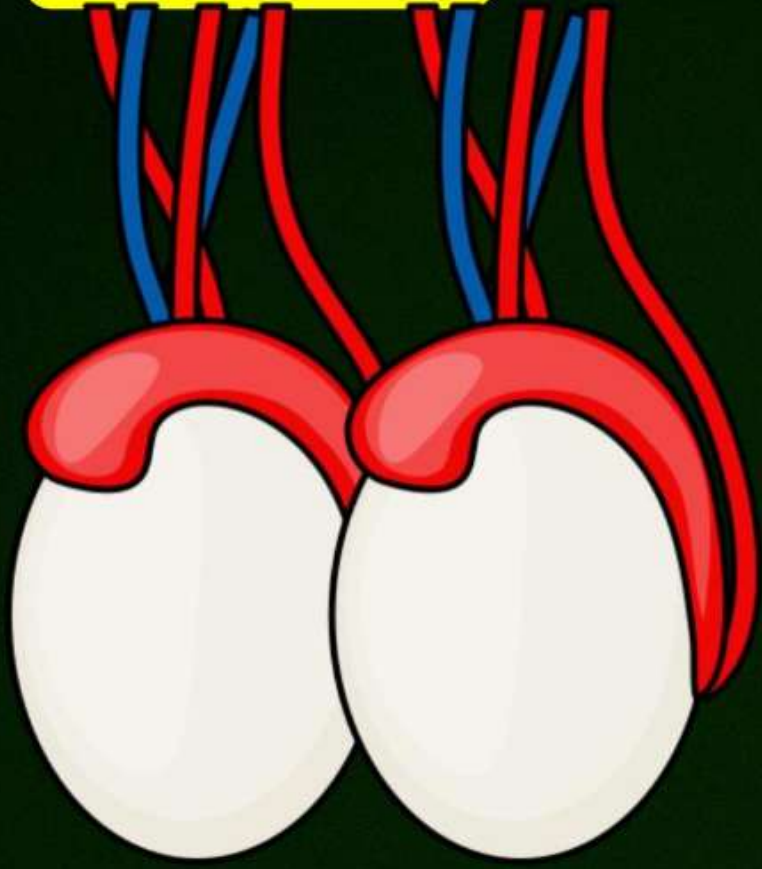


Physics Wallah

Topics To Be Covered

- 1 **ENDOCINE GLANDS** → Testes
→ ovaries
- 2 **HORMONES OF HEART, KIDNEY & GI TRACT**
- 3 **MECHANISM OF HORMONE ACTION**
- 4 **TYPES OF HORMONES**
- 5 **PREVIOUS & CURRENT YEAR NEET MCQ's**

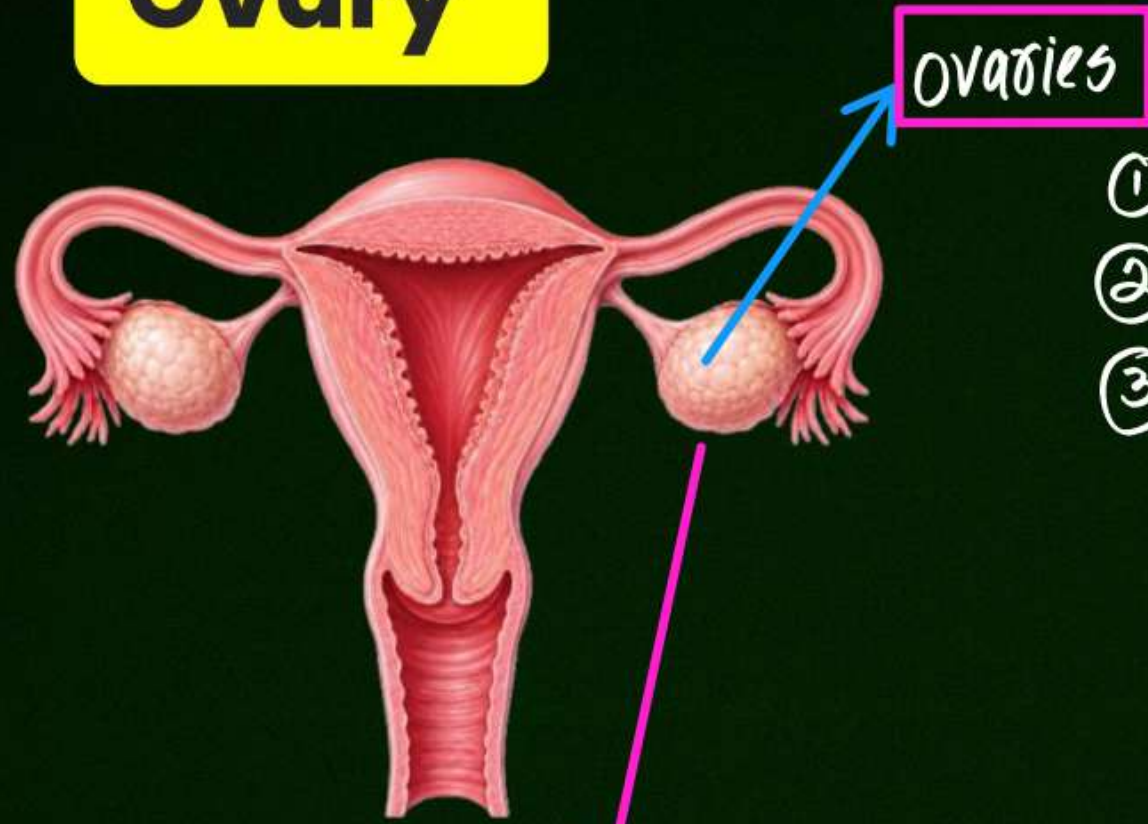
Testes



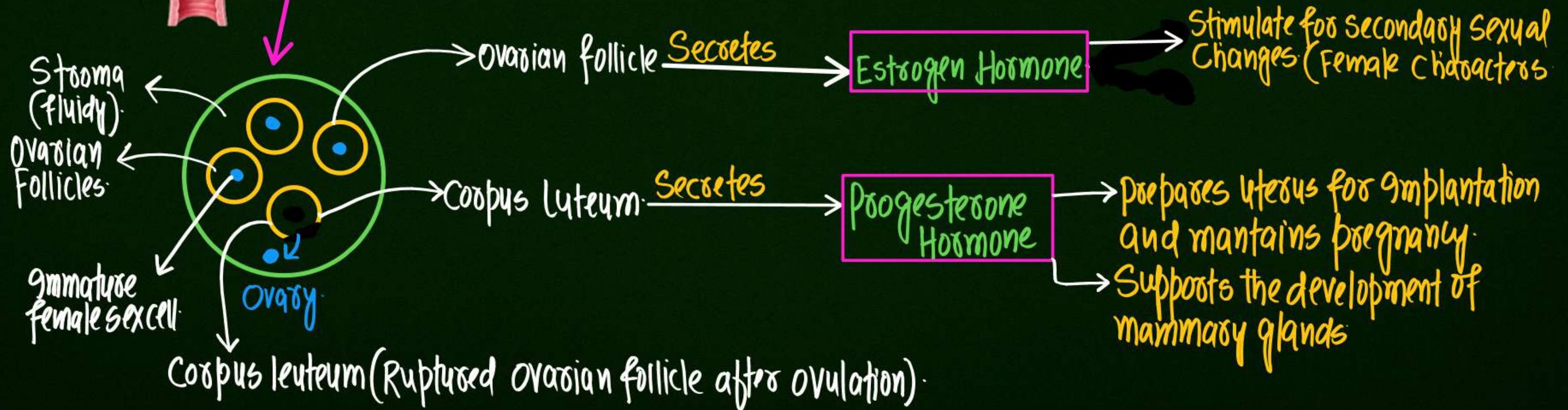
- ① present in Sacrotum of Males.
- ② they are primary Sex-organs i.e they produce Sperm.
- ③ they also function as Endocrine gland i.e they produce Hormone



Ovary



- ① Present in abdominal cavity of females
- ② they are the primary sex organs in females i.e they produce sex cell.
- ③ they also act as Endocrine gland i.e they produce hormones.

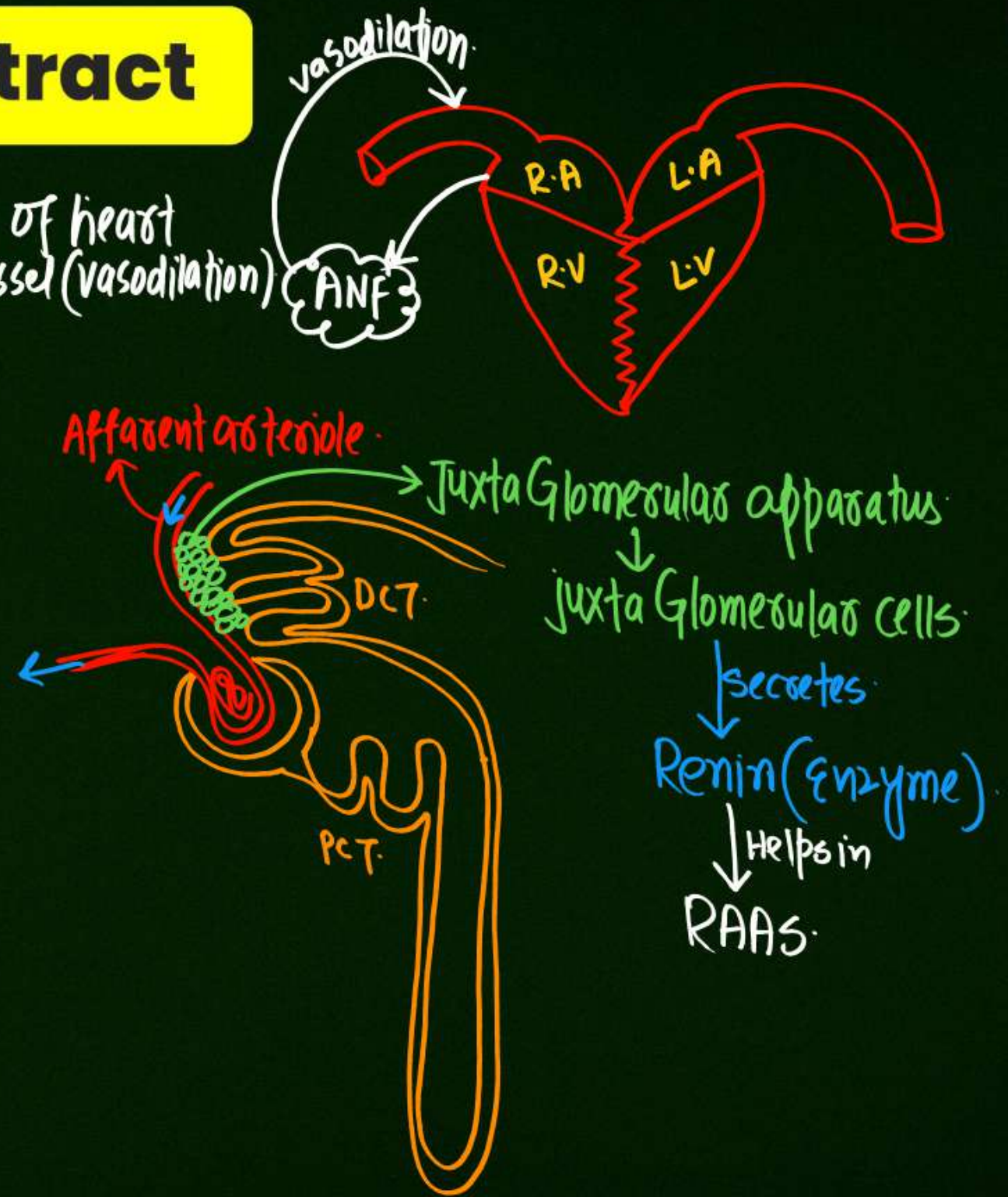


Hormones of Heart, Kidney & GI tract

Heart: → Atrial natriuretic factor (ANF) is secreted by atria of heart when blood pressure is high and it dilates blood vessel (vasodilation).

Kidney: → Renin is secreted from JG-cells.
→ Erythropoietin is secreted by kidney which helps in erythropoiesis (RBC production).

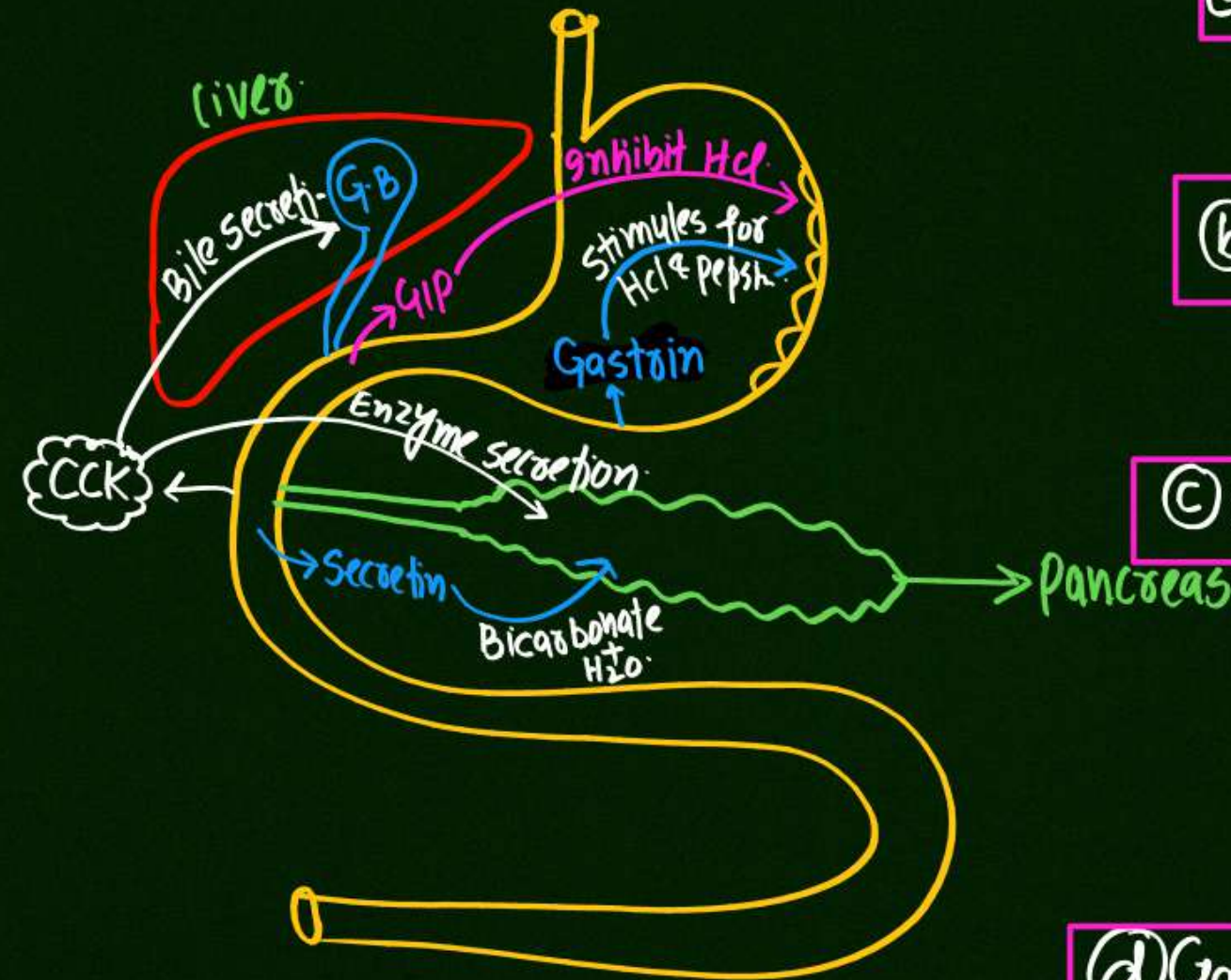
Gastrointestinal Tract (GI Tract):



Hormones of Heart, Kidney & GI tract



GI tract



a) Gastrin → Secreted by antrum of stomach and stimulates gastric glands for HCl and pepsinogen production.

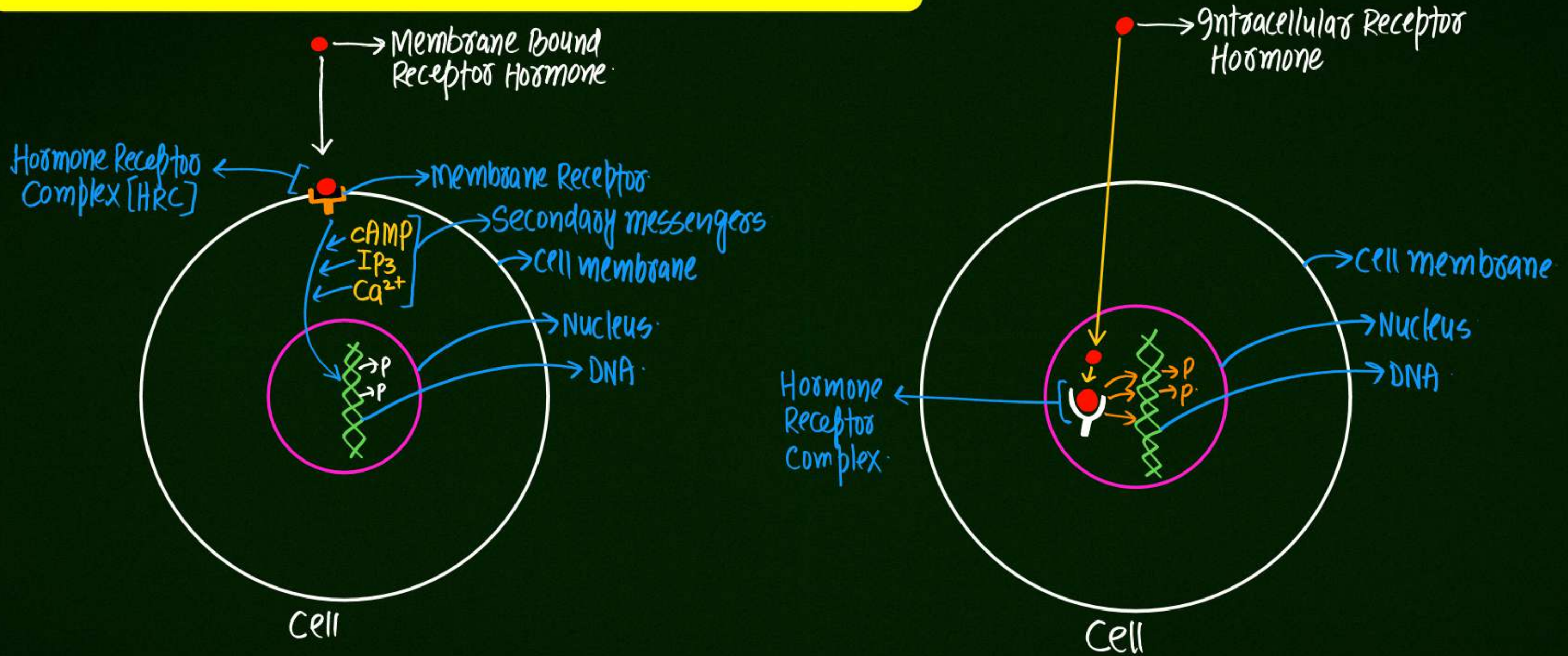
b) Secretin → Secreted by duodenum and stimulates pancreas for bicarbonate and H₂O production.

c) Cholecystokinin (CCK) → Secreted by duodenum and stimulates gall bladder for bile secretion and pancreas for enzyme secretion.

d) Gastric inhibitory peptide → Secreted from duodenum and inhibits gastric secretion and motility.

Other Non-endocrine tissues secrete growth factors essential for tissue growth and regeneration.

Mechanism of Hormone Action



Mechanism of Hormone Action

Hormones act on target tissues by binding to specific hormone receptors & forms a hormone-receptor complex which triggers biochemical changes in the target tissue. Each receptor is specific to a hormone & cells contain two types of receptors:

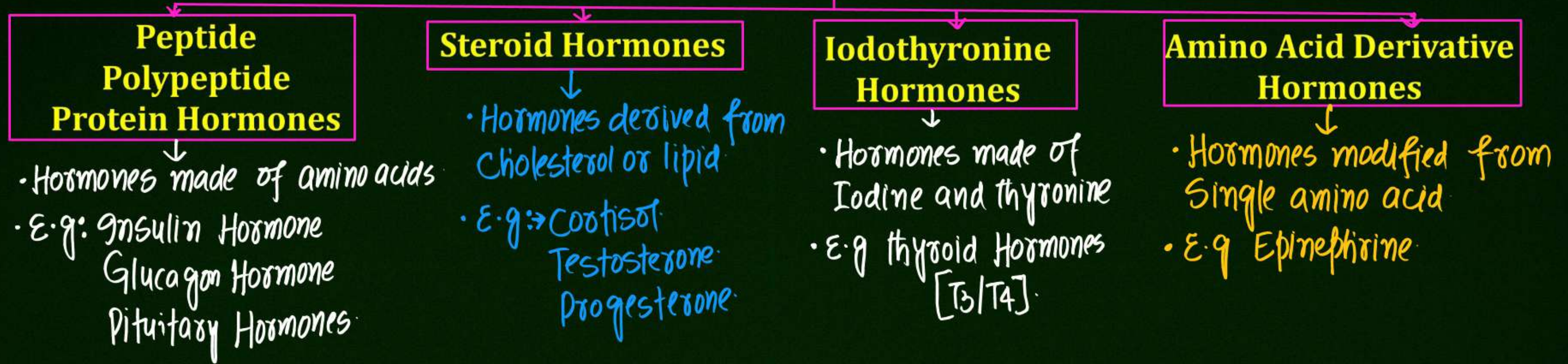
- a) Membrane-bound receptors (on cell surface)
- b) Intracellular receptors (usually nuclear)

Receptor Interaction Types:

- a) Membrane-bound receptor hormones: Hormones bind to receptor present on cell & do not enter the cell & use second messengers like (e.g., cAMP, IP₃, Ca²⁺) to affect metabolism.
- b) Intracellular receptor hormones (e.g., steroids, iodothyronines): Hormones enter the cell & bind to receptors present on nucleus & regulate gene expression by interacting with DNA. Lead to physiological and developmental effects.

Types of Hormone

→ on the basis of chemical nature hormones are of four types:



Kaun Banayga Nabkaar [NEET-2025]

Match List I with List II:

List I

List II

- | | | |
|-----------------------------|---|--------------------------|
| (A) Heart | → | (I) Erythropoietin |
| (B) Kidney | → | (II) Aldosterone |
| (C) Gastro-intestinal tract | → | (III) Atrial natriuretic |
| (D) Adrenal Cortex | → | (IV) Secretin |

Choose the correct answer from the options given below:

A

A-IV, B-III, C-II, D-I ✗

B

A-I, B-III, C-IV, D-II ✗

C

A-III, B-I, C-IV, D-II ✓

D

A-II, B-I, C-III, D-IV ✗

CLASS 11TH (BAAZ 1.0)
SYLLABUS COMPLETED.
61-LECTURES (100 HOURS).
244-NEET MCQ's SOLVED

Thank
You